

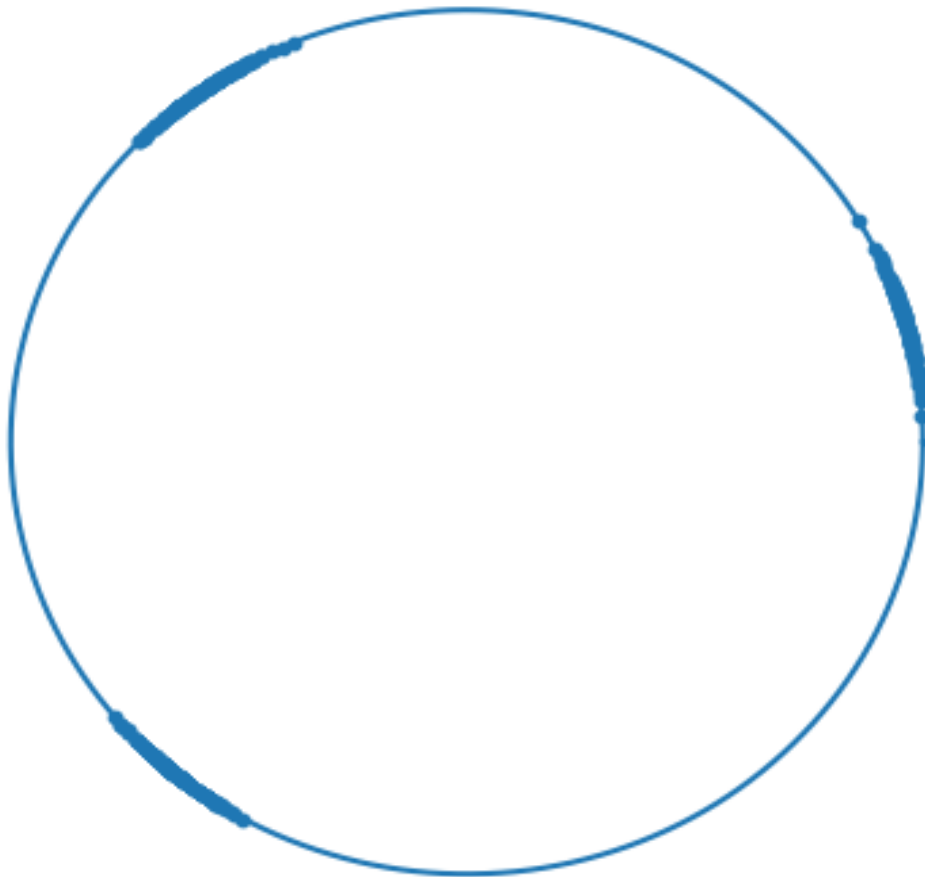
Angular Manifold Routing — Visual Explainer (Version 3)

This document explains a geometric routing method that may significantly improve the efficiency of large AI models. The key idea is simple: when token embeddings are normalized and routed by their direction on a hypersphere, tokens naturally cluster into fewer compute groups than expected.

1. Token Clustering in Direction Space

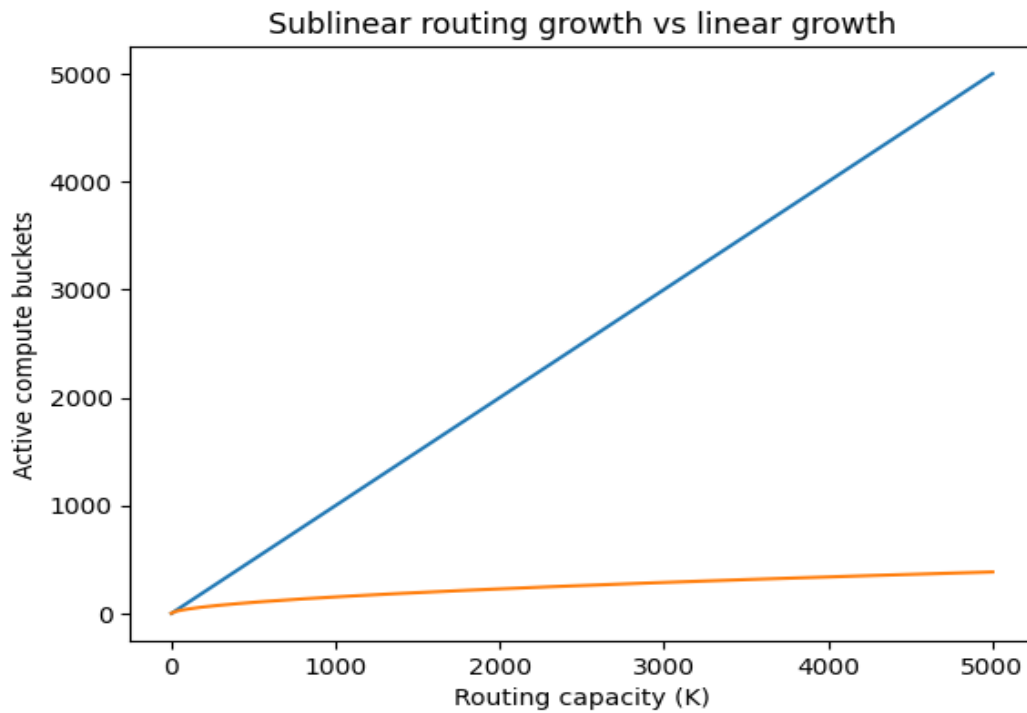
Embeddings representing related concepts tend to point in similar directions in vector space. After normalization, embeddings lie on the surface of a hypersphere. Routing based on direction causes tokens to cluster naturally into angular sectors.

Token embeddings clustering by direction



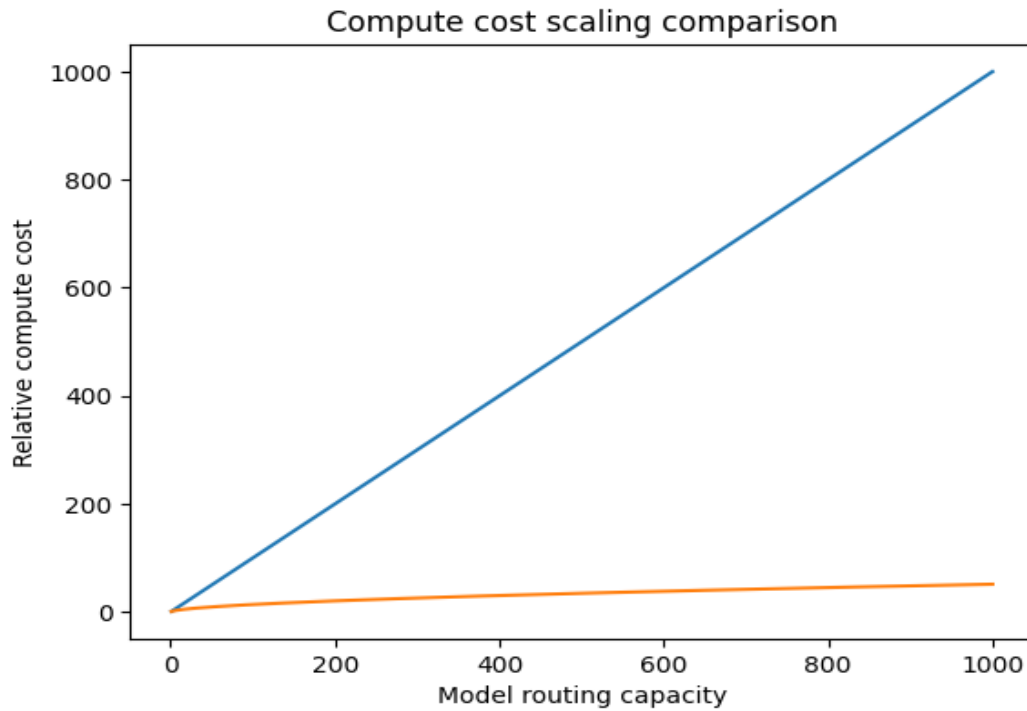
2. Sublinear Routing Growth

Experiments revealed that the number of active routing buckets grows sublinearly with routing capacity. The observed relationship is approximately: $E(K) \approx c \cdot K^{0.57}$ This means doubling routing capacity does not double compute usage. Instead, compute grows much more slowly.



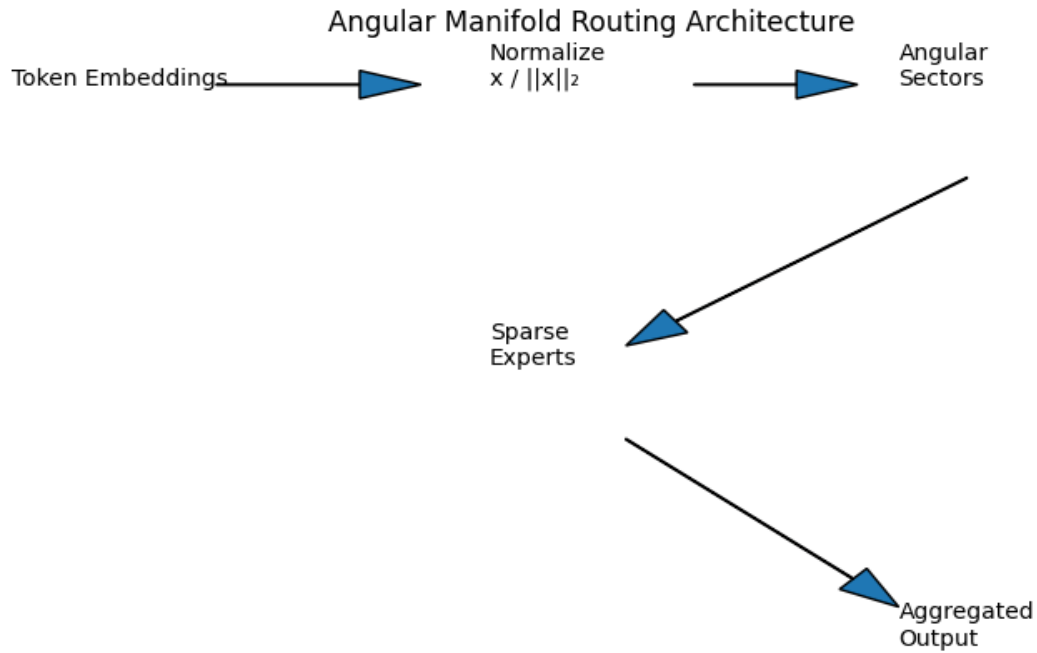
3. Hardware Scaling Implications

Because routing buckets grow sublinearly, the amount of compute required to support larger routing capacity grows much more slowly than traditional architectures. This effect can translate into substantial hardware efficiency gains.



4. Architecture Overview

The routing architecture can be summarized as a simple pipeline: 1. Tokens are converted into embeddings. 2. Embeddings are normalized onto a unit hypersphere. 3. The hypersphere is divided into angular sectors. 4. Tokens are routed to sectors based on direction. 5. Only active sectors perform computation. 6. Outputs are aggregated into the model output. This design removes the need for complex gating networks and allows the geometry of the embedding space to organize computation automatically.



Summary

Angular Manifold Routing suggests that geometric structure in embedding space can naturally produce sparse computation. When tokens are routed by direction on a normalized hypersphere, they cluster into fewer compute groups than expected. This effect could enable more scalable AI systems with lower hardware requirements.